

# MODULE – II COMPLETE ANSWERS (DBMS)

Based on the uploaded question bank.

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## SHORT ANSWER QUESTIONS (5 MARKS)

### 1. Explain Derived Attribute and Multivalued Attribute with Example and Diagram

#### Derived Attribute

A derived attribute is an attribute whose value can be calculated from another attribute.

#### Example

If Date\_of\_Birth is stored, then Age can be calculated.

Student

-----

DOB

Age

Age is a derived attribute.

#### ER Notation

Derived attributes are represented by a **dashed ellipse**.

(Age)

-----

|

Student

### Example

DOB = 01-01-2005

Current Year = 2025

Age = 20

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## Multivalued Attribute

An attribute that can have multiple values for a single entity.

### Example

An employee can have multiple phone numbers.

Employee

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EmpID

Name

Phone Numbers

### ER Notation

Represented using a **double ellipse**.

((Phone))

|

Employee

### Example

Employee 101

Phone:

9876543210

9123456780

9988776655

---

## 2. What is Key Attribute? Explain with Example

A key attribute uniquely identifies an entity in an entity set.

### Example

Student Entity

| <b>SID</b> | <b>Name</b> | <b>Dept</b> |
|------------|-------------|-------------|
| 101        | Anik        | CSE         |
| 102        | Rahul       | EE          |

SID uniquely identifies each student.

Therefore SID is the key attribute.

### ER Notation

Student

-----

SID

Name

Dept

SID is underlined.

Student

-----

SID

Name

Dept

### Importance

- Uniqueness
  - Prevents duplicate records
  - Supports relationships
-

### 3. What is Foreign Key? Explain with Example

A foreign key is an attribute in one table that refers to the primary key of another table.

#### Example

Department Table

| DeptID | DeptName |
|--------|----------|
| 10     | CSE      |
| 20     | EE       |

Employee Table

| EmpID | Name | DeptID |
|-------|------|--------|
| 1     | Ravi | 10     |
| 2     | Anik | 20     |

DeptID in Employee table is a Foreign Key.

#### SQL

```
CREATE TABLE Department(  
  DeptID INT PRIMARY KEY,  
  DeptName VARCHAR(20)  
);
```

```
CREATE TABLE Employee(  
  EmpID INT PRIMARY KEY,  
  Name VARCHAR(20),  
  DeptID INT,  
  FOREIGN KEY(DeptID)  
  REFERENCES Department(DeptID)  
);
```

#### Purpose

- Maintains referential integrity
- Connects tables

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## 4. Explain Hierarchical Data Model with Example

Hierarchical Model organizes data in a tree structure.

### Features

- Parent-child relationship
- One parent can have many children
- Child has only one parent

### Example

Company  
|  
Departments  
|  
Employees

### Diagram

Company  
|  
-----  
|            |  
CSE        EE  
|            |  
Emp1       Emp3  
Emp2       Emp4

### Advantages

- Fast access
- Simple structure

### Disadvantages

- Difficult relationships
  - Poor flexibility
-

## 5. Define Entity, Attribute, Entity Set, Relationship with Appropriate Notations

### Entity

A real-world object distinguishable from others.

#### Example

Student

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### Attribute

Property describing an entity.

#### Example

Name, Age, RollNo

---

### Entity Set

Collection of similar entities.

#### Example

All students in a college.

---

### Relationship

Association among entities.

#### Example

Student enrolls in Course.

---

## ER Notations

### Entity

Rectangle

```
+-----+  
| Student |  
+-----+
```

### Attribute

Ellipse

(Name)

### Relationship

Diamond

```
Student  
  |  
Enrolls  
  |  
Course
```

---

## Multivalued Attribute Notation

((Phone))

(Double ellipse)

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## 6. Draw ER Diagram for Ternary Relationship Set with Example

A ternary relationship involves three entity sets simultaneously.

### Example

Supplier supplies Part to Project.

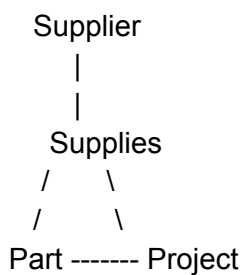
Entities:

- Supplier
- Part
- Project

Relationship:

- Supplies

### ER Diagram



### Example

Supplier S1 supplies Part P1 to Project PR1.

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## 7. What Does Cardinality Ratio Specify? List Eight Applications of DBMS

### Cardinality Ratio

Specifies number of entities participating in a relationship.

### Types

#### One-to-One (1:1)

Person ↔ Passport

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#### One-to-Many (1:M)

Department ↔ Employees

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### **Many-to-One (M:1)**

Employees ↔ Department

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### **Many-to-Many (M:N)**

Students ↔ Courses

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## **Eight Applications of DBMS**

1. Banking Systems
  2. Railway Reservation
  3. Airline Reservation
  4. Library Management
  5. Hospital Management
  6. E-Commerce
  7. University Systems
  8. Telecommunications
- 

## **8. Define Relational Database. Explain Data Representation in RDBMS**

A Relational Database stores data in the form of tables.

### **Example**

Student Table

| <b>SID</b> | <b>Name</b> | <b>Dept</b> |
|------------|-------------|-------------|
| 101        | Anik        | CSE         |

---

# Data Representation

## Relation

Table

## Tuple

Row

## Attribute

Column

## Domain

Permissible values

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## Example

Relation = STUDENT

Attributes:

SID

NAME

DEPT

Tuple:

(101, Anik, CSE)

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## 9. What is Referential Integrity Constraint? How is it Related to Foreign Key?

Referential Integrity ensures every foreign key value exists in the referenced table.

## Example

Department

**DeptID**

10

20

Employee

**EmpID    DeptID**

1            10

Valid because 10 exists.

---

Invalid:

**EmpID    DeptID**

1            50

Since DeptID 50 doesn't exist.

## **Relationship**

Foreign Key implements Referential Integrity.

## **Benefits**

- Prevents orphan records
  - Maintains consistency
- 

# **LONG ANSWER QUESTIONS (10 MARKS)**

## **1. Define Weak Entity Set, Candidate Key, Primary Key and Foreign Key**

## Weak Entity Set

An entity that cannot be uniquely identified without another entity.

Depends on a strong entity.

### Example

Employee → Dependent

Dependent cannot be identified independently.

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## ER Notation

Double Rectangle

```
+=====+  
| Dependent |  
+=====+
```

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## Candidate Key

A minimal set of attributes that uniquely identifies a tuple.

### Example

Student

SID  
Email

Both may uniquely identify students.

Candidate Keys:

SID  
Email

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## Primary Key

One candidate key selected as the main identifier.

Example:

SID

---

## Foreign Key

Attribute referencing primary key of another relation.

Example:

Employee(DeptID)

Department(DeptID)

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## Comparison

| Key Type      | Purpose                |
|---------------|------------------------|
| Candidate Key | Possible identifier    |
| Primary Key   | Chosen identifier      |
| Foreign Key   | Establish relationship |

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## 2. Explain Generalization and Specialization with Example

### Generalization

Combining lower-level entities into a higher-level entity.

#### Example

Car  
Truck  
Bus

Generalized into

Vehicle

## Diagram

Car  
Truck  
Bus  
|  
Vehicle

---

## Specialization

Breaking a higher-level entity into sub-entities.

## Example

Employee

Specialized into:

Manager  
Engineer  
Clerk

## Diagram

Employee  
|  
-----  
| | |  
Mgr Eng Clerk

---

## Difference

### Generalization

Bottom-up

Merge entities

### Specialization

Top-down

Divide entities

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### 3. Difference Between Hierarchical and Network Model

| Feature      | Hierarchical | Network          |
|--------------|--------------|------------------|
| Structure    | Tree         | Graph            |
| Parent       | One Parent   | Multiple Parents |
| Relationship | One-to-Many  | Many-to-Many     |
| Complexity   | Simple       | Complex          |
| Flexibility  | Less         | More             |

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#### Hierarchical Example

Company  
|  
Department  
|  
Employee

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#### Network Example

Student  
/ \  
Course Teacher

A student can connect to multiple courses.

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### 4. Explain Integrity Constraints Over Relations

Integrity constraints ensure correctness and consistency.

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## 1. Domain Constraint

Values must belong to a domain.

Example:

Age > 0

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## 2. Key Constraint

Primary key must be unique.

Example:

SID cannot repeat.

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## 3. Entity Integrity

Primary key cannot be NULL.

Example:

SID cannot be NULL.

---

## 4. Referential Integrity

Foreign key must reference existing primary key.

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### Example

Department(DeptID)

Employee(EmpID, DeptID)

DeptID must exist in Department.

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## 5. ER Diagram for Company Database

Entities:

### **Employee**

Attributes

SSN (PK)

Salary

Phone

### **Department**

Attributes

DNO (PK)

DName

Budget

### **Child (Weak Entity)**

Attributes

Name

Age

---

Relationships

### **Works\_In**

Employee → Department

(Many : One)

### **Manages**

Employee → Department

(One : One)

## Has

Employee → Child

(One : Many)

---

## ER Diagram

Employee

(SSN, Salary, Phone)

|

Works\_In

|

Department

(DNO,DName,Budget)

Employee

|

Has

|

Child(Name,Age)

Employee

|

Manages

|

Department

Child is weak because it depends on Employee.

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## 6. ER Diagram for Company–Department–Employee–Dependent

Requirements:

- Company has four departments.
- Department belongs to one company.
- Department employs many employees.
- Employee works for one department.

- Employee may have dependents.

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## Entities

Company

CID  
CName

Department

DID  
DName

Employee

EID  
Name

Dependent

DepID  
Name

---

## Relationships

Company

|  
Has

|  
Department

|  
Employs

|  
Employee

|  
Has

|  
Dependent

## Cardinalities

Company : Department = 1 : 4

Department : Employee = 1 : N

Employee : Dependent = 1 : N

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## **MODULE II SUMMARY FOR EXAM**

Important topics frequently asked:

1. Entity, Attribute, Entity Set, Relationship
2. Key Attribute
3. Foreign Key
4. Referential Integrity
5. Hierarchical vs Network Model
6. Generalization and Specialization
7. Weak Entity Set
8. Cardinality Ratio
9. ER Diagrams
10. Integrity Constraints

These topics alone cover a large portion of Module II questions.